

Put Option Buying

The purchase of a put option provides leverage in the case of a downward move by the underlying stock. In this manner, *it is an alternative to the short sale of stock*, much as the purchase of a call option is a leveraged alternative to the purchase of stock.

PUT BUYING VERSUS SHORT SALE

In the simplest case, when an investor expects a stock to decline in price, he may either short the underlying stock or buy a put option on the stock. Suppose that XYZ is at 50 and that an XYZ July 50 put option is trading at 5. If the underlying stock declines substantially, *the buyer of the put could make profits considerably in excess of his initial investment*. However, if the underlying stock rises in price, *the put buyer has limited risk*; he can lose only the amount of money that he originally paid for the put option. In this example, the most that the put buyer could lose would be 5 points, which is equal to his entire initial investment. Table 16-1 and Figure 16-1 depict the results, at expiration, of this simple purchase of the put option.

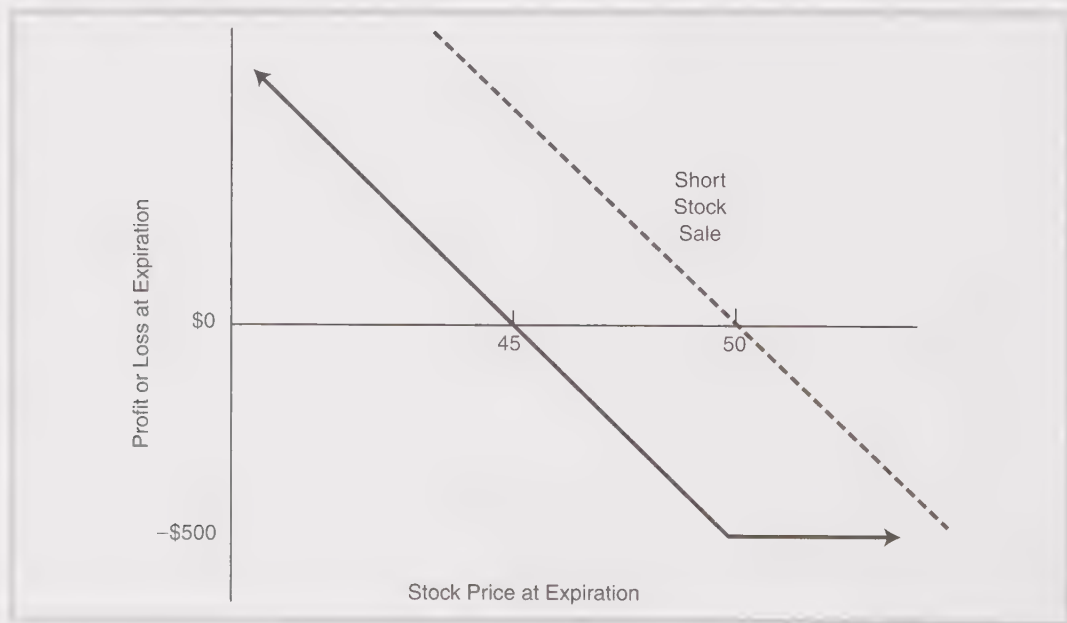
The put buyer has limited profit potential, since a stock can never drop in price below zero dollars per share. However, his potential profits can be huge, percentagewise. His loss, which normally would occur if the stock rises in price, is limited to the amount of his initial investment. *The simplest use of a put purchase is for speculative purposes when expecting a price decline in the underlying stock.*

These results for the profit or loss of the put option purchases can be compared to a similar short sale of XYZ at 50 in order to observe the benefits of leverage and limited risk that the put option buyer achieves. In order to sell short 100 XYZ at 50, assume that the trader would have to use \$2,500 in margin. Several points can be verified from Table 16-2 and Figure 16-1. If the stock drops in price sufficiently far, the percentage profits are

TABLE 16-1.
Results of put purchase at expiration.

XYZ Price at Expiration	Put Price at Expiration	Put Option Profit
20	30	+\$2,500
30	20	+ 1,500
40	10	+ 500
45	5	0
48	2	- 300
50	0	- 500
60	0	- 500
70	0	- 500

FIGURE 16-1.
Put option purchase.



much greater on the put option purchase than they are on the short sale of the underlying stock. This is the leveraging effect that an option purchase can achieve. If the underlying stock remains relatively unchanged, the short seller would do better because he does not risk losing his entire investment in a limited amount of time if the underlying stock

TABLE 16-2.
Results of selling short.

XYZ Price at Expiration	Short Sale	Put Option Purchase
20	+\$3,000 (+120%)	+\$2,500 (+ 500%)
30	+ 2,000 (+ 80%)	+ 1,500 (+ 300%)
40	+ 1,000 (+ 40%)	+ 500 (+ 100%)
45	+ 500 (+ 20%)	0 (0%)
48	+ 200 (+ 8%)	– 300 (– 60%)
50	0 (0%)	– 500 (– 100%)
60	– 1,000 (– 40%)	– 500 (– 100%)
75	– 2,500 (–100%)	– 500 (– 100%)
100	– 5,000 (–200%)	– 500 (– 100%)

changes only slightly in price. However, if the underlying stock should rise dramatically, the short seller can actually lose more than his initial investment. The short sale of stock has theoretically unlimited risk. Such is not true of the put option purchase, whereby the risk is limited to the amount of the initial investment. One other point should be made when comparing the purchase of a put and the short sale of stock. The short seller of stock is obligated to pay the dividends on the stock, but the put option holder has no such obligation. This is an additional advantage to the holder of the put.

SELECTING WHICH PUT TO BUY

Many of the same types of analyses that the call buyer goes through in deciding which call to buy can be used by the prospective put buyer as well. First, when approaching put buying as a speculative strategy, one should not place more than 15% of his risk capital in the strategy. Some investors participate in put buying to add some amount of downside protection to their basically bullishly oriented common stock portfolios. More is said in Chapter 17 about buying puts on stocks that one actually owns.

The out-of-the-money put offers both higher reward potentials and higher risk potentials than does the in-the-money put. If the underlying stock drops substantially, the percentage returns from having purchased a cheaper, out-of-the-money put will be greater. However, should the underlying stock decline only moderately in price, the *in-the-money put* will often prove to be the better choice. In fact, since a put option tends

to lose its time value premium quickly as it becomes an in-the-money option, there is an even greater advantage to the purchase of the in-the-money put.

Example: XYZ is at 49 and the following prices exist:

XYZ, 49;

XYZ July 45 put, 1; and

XYZ July 50 put, 3.

If the underlying stock were to drop to 40 by expiration, the July 45 put would be worth 5 points, a 400% profit. The July 50 put would be worth 10 points, a 233% profit over its initial purchase price of 3. Thus, in a substantial downward move, the out-of-the-money put purchase provides higher reward potential. However, if the underlying stock drops only moderately, say to 45, the purchaser of the July 45 put would lose his entire investment, since the put would be worthless at expiration. The purchaser of the in-the-money July 50 put would have a 2-point profit with XYZ at 45 at expiration.

The preceding analysis is based on holding the put until expiration. For the option buyer, this is generally an erroneous form of analysis, because the buyer generally tends to liquidate his option purchase in advance of expiration. When considering what happens to the put option in advance of expiration, it is helpful to remember that an in-the-money put tends to lose its time premium rather quickly. In the example above, the July 45 put is completely composed of time value premium. If the underlying stock begins to drop below 45, the price of the put will not increase as rapidly as would the price of a call that is going into-the-money.

Example: If XYZ fell by 5 points to 44, definitely a move in the put buyer's favor, he may find that the July 45 put has increased in value only to 2 or 2½ points. This is somewhat disappointing because, with call options, one would expect to do significantly better on a 5-point stock movement in his favor. Thus, when purchasing put options for speculation, *it is generally best to concentrate on in-the-money puts unless a very substantial decline in the price of the underlying stock is anticipated.*

Once the put option is in-the-money, the time value premium will decrease even in the longer-term series. Since this time premium is small in all series, the put buyer can often purchase a longer-term option for very little extra money, thus gaining more time to work with. Call option buyers are generally forced to avoid the longer-term series because the extra cost is not worth the risk involved, especially in a trading situation. However, the put buyer does not necessarily have this disadvantage. If he can purchase the longer-term

put for nearly the same price as the near-term put, he should do so in case the underlying stock takes longer to drop than he had originally anticipated it would.

It is not uncommon to see such prices as the following:

XYZ common, 46;

XYZ April 50 put, 4;

XYZ July 50 put, 4.50; and

XYZ October 50 put, 5.

None of these three puts have much time value premium in their prices. Thus, the buyer might be willing to spend the extra 1 point and buy the longest-term put. If the underlying stock should drop in price immediately, he will profit, but not as much as if he had bought one of the less expensive puts. However, should the underlying stock rise in price, he will own the longest-term put and will therefore suffer less of a loss, percentage-wise. If the underlying stock rises in price, some amount of time value premium will come back into the various puts, and the longest-term put will have the largest amount of time premium. For example, if the stock rises back to 50, the following prices might exist:

XYZ common, 50;

XYZ April 50 put, 1;

XYZ July 50 put, 2.50; and

XYZ October 50 put, 3.50.

The purchase of the longer-term October 50 put would have suffered the least loss, percentage-wise, in this event. Consequently, when one is purchasing an in-the-money put, he may often want to consider buying the longest-term put if the time value premium is small when compared to the time premium in the nearer-term puts.

In Chapter 3, the delta of an option was described as the amount by which one might expect the option will increase or decrease in price if the underlying stock moves by a fixed amount (generally considered to be one point, for simplicity). Thus, if XYZ is at 49 and a call option is priced at 3 with a delta of .50, one would expect the call to sell for 3.50 with XYZ at 50 and to sell at 2.50 with XYZ at 48. In reality, the delta changes even on a fractional move in the underlying stock, but one generally assumes that it will hold true for a 1-point move. Obviously, put options have deltas as well. The delta of a put is a negative number, reflecting the fact that the put price and the stock price are inversely related. *As an approximation, one could say that the delta of the call option minus the delta of the put option with the same terms is equal to 1.* That is,

$$\text{Delta of put} = \text{Delta of call} - 1.$$

This is an approximation and is accurate unless the put is deeply in-the-money. It has already been pointed out that the time value premium behavior of puts and calls is different, so it is inaccurate to assume that this formula holds true exactly for all cases.

The delta of a put ranges between 0 and minus 1. If a July 50 put has a delta of $-.50$, and the underlying stock *rises* by 1 point, the put will lose $.50$. The delta of a deeply out-of-the-money put is close to zero. The put's delta would decrease slowly at first as the stock declined in value, then would begin to decrease much more rapidly as the stock fell through the striking price, and would reach a value of minus 1 (the minimum) as the stock fell only moderately below the striking price. This is reflective of the fact that an out-of-the-money put tends to hold time premium quite well and an in-the-money put comes to parity rather quickly.

RANKING PROSPECTIVE PUT PURCHASES

In Chapter 3, a method of ranking prospective call purchases was developed that encompassed certain factors, such as the volatility of the underlying stock and the expected holding period of the purchased option. The same sort of analysis should be applied to put option purchases.

The steps are summarized below. The reader may refer to the section titled "Advanced Selection Criteria" in Chapter 3 for a more detailed description of why this method of ranking is superior.

1. Assume that each underlying stock can decrease in price in accordance with its volatility over a fixed holding period (30, 60, or 90 days).
2. Estimate the put option prices after the decrease.
3. Rank all potential put purchases by the highest reward opportunity for aggressive purchases.
4. Estimate how much would be lost if the underlying stock instead rose in accordance with its volatility, and rank all potential put purchases by best risk/reward ratio for a more conservative list of put purchases.

As was stated earlier, it is necessary to have a computer to make an accurate analysis of all listed options. The average customer is forced to obtain such data from a brokerage firm or data service. He should be sure that the list he is using conforms to the above-mentioned criteria. If the data service is ranking option purchases by how well the puts would do if each underlying stock fell by a fixed percentage (such as 5% or 10%), the list should be rejected because it is not incorporating the volatility of the underlying stock into its analysis. Also, if the list is based on holding the put purchase until expiration, the

list should be rejected as well, because this is not a realistic assumption. There are enough reliable and sophisticated data services that one should not have to work with inferior analyses in today's option market.

For those readers who are more mathematically advanced and have the computer capability to construct their own analyses, the details of implementing an analysis similar to the one described above are presented in Chapter 28, *Mathematical Applications*. An application of put purchases, combined with fixed-income securities, is described in Chapter 26, *Buying Options and Treasury Bills*.

FOLLOW-UP ACTION

The put buyer can take advantage of strategies that are very similar to those the call buyer uses for follow-up action, either to lock in profits or to attempt to improve a losing situation. Before discussing these specific strategies, it should be stated again that it is rarely to the option buyer's benefit to exercise the option in order to liquidate. This precludes, of course, those situations in which the call buyer actually wants to own the stock or the put buyer actually wants to sell the stock. If, however, the option holder is merely looking to liquidate his position, the cost of stock commissions makes exercising a prohibitive move. This is true even if he has to accept a price that is a slight discount from parity when he sells his option.

LOCKING IN PROFITS

The reader may recall that there were four strategies (perhaps "tactics" is a better word) for the call buyer with an unrealized profit. These same four tactics can be used with only slight variations by the put option buyer. Additionally, a fifth strategy can be employed when a stock has both listed puts and calls.

After an underlying stock has moved down and the put buyer has a relatively substantial unrealized gain, he might consider taking one of the following actions:

1. Sell the put and liquidate the position for a profit.
2. Do nothing and continue to hold the put.
3. Sell the in-the-money long put and use part of the proceeds to purchase out-of-the-money puts.
4. Create a spread by selling an out-of-the-money put against the one he currently holds.

These are the same four tactics that were discussed earlier with respect to call buying. In the fifth tactic, the holder of a listed put who has an unrealized profit might consider buying a listed *call* to protect his position.

TABLE 16-3.
Background table for profit alternatives.

Original Trade	Current Prices	
XYZ common: 52	XYZ common:	45
Bought XYZ October 50 put at 2	XYZ October 50 put:	6
	XYZ October 45 put:	2

Example: A speculator originally purchased an XYZ October 50 put for 2 points when the stock was 52. If the stock has now fallen to 45, the put might be worth 6 points, representing an unrealized gain of 4 points and placing the put buyer in a position to implement one of these five tactics. After some time has passed, with the stock at 45, an at-the-money October 45 put might be selling for 2 points. Table 16-3 summarizes the situation. If the trader merely liquidates his position by selling out the October 50 put, he would realize a profit of 4 points. Since he is terminating the position, he can make neither more nor less than 4 points. This is the most conservative of the tactics, allowing no additional room for appreciation, but also eliminating any chance of losing the accumulated profits.

If the trader does nothing, merely continuing to hold the October 50 put, he is taking an aggressive action. If the stock should reverse and rise back above 50 by expiration, he would lose everything. However, if the stock continues to fall, he could build up substantially larger profits. This is the only tactic that could eventually result in a loss at expiration.

These two simple strategies—liquidating or doing nothing—are the easiest alternatives. The remaining strategies allow one to attempt to achieve a balance between retaining built-up profits and generating even more profits. The third tactic that the speculator could use would be to sell the put that he is currently holding and use some of the proceeds to purchase the October 45 put. *The general idea in this tactic is to pull one's initial investment out of the market* and then to increase the number of option contracts held by buying the out-of-the-money option.

Example: The trader would receive 6 points from the sale of the October 50 put. He should take 2 points of this amount and put it back into his pocket, thus covering his initial investment. Then he could buy 2 October 45 puts at 2 points each with the remaining portion of the proceeds from the sale. He has no risk at expiration with this strategy, since he has recovered his initial investment. Moreover, if the underlying stock should continue to fall rapidly, he could profit handsomely because he has increased the number of put contracts that he holds.

The fourth choice that the put holder has is to create a spread by selling the October 45 put against the October 50 that he currently holds. This would create a bear spread,

technically. This type of spread is described in more detail later. For the time being, it is sufficient to understand what happens to the trader's risks and rewards by creating this spread. The sale of the October 45 put brings in 2 points, which covers the initial 2-point purchase cost of the October 50 put. Thus, *his "cost" for this spread is nothing*; he has no risk, except for commissions. If the underlying stock should rise above 50 by expiration, all the puts would expire worthless. (A put expires worthless when the underlying stock is above the striking price at expiration.) This would represent the worst case; he would recover nothing from the spread. If the stock should be below 45 at expiration, he would realize the maximum potential of the spread, which is 5 points. That is, no matter how far XYZ is below 45 at expiration, the October 50 put will be worth 5 points more than the October 45 put, and the spread could thus be liquidated for 5 points. His maximum profit potential in the spread situation is 5 points. This tactic would be the best one if the underlying stock stabilized near 45 until expiration.

To analyze the fifth strategy that the put holder could use, it is necessary to introduce a call option into the picture.

Example: With XYZ at 45, there is an October 45 call selling for 3 points. The put holder could buy this call in order to limit his risk and still retain the potential for large future profits. If the trader buys the call, he will have the following position:

Long 1 October 50 put
Long 1 October 45 call – Combined cost: 5 points

The total combined cost of this put and call combination is 5 points—2 points were originally paid for the put, and now 3 points have been paid for the call. No matter where the underlying stock is at expiration, this combination will be worth at least 5 points. For example, if XYZ is at 46 at expiration, the put will be worth 4 and the call worth 1; or if XYZ is at 48, the put will be worth 2 and the call worth 3. If the stock is above 50 or below 45 at expiration, the combination will be worth more than 5 points. Thus, the trader has no risk in this combination, since he has paid 5 points for it and will be able to sell it for at least 5 points at expiration. In fact, if the underlying stock continues to drop, the put will become more valuable and he could build up substantial profits. Moreover, if the underlying stock should reverse direction and climb substantially, he could still profit, because the call will then become valuable. This tactic is the best one to use if the underlying stock does not stabilize near 45, but instead makes a relatively dramatic move either up or down by expiration. The strategy of simultaneously owning both a put and a call is discussed in much greater detail in Chapter 23. It is introduced here merely for the purposes of the put buyer wanting to obtain protection of his unrealized profits.

Each of these five strategies may work out to be the best one under a different set of circumstances. The ultimate result of each tactic is dependent on the direction that XYZ moves in the future. As was the case with call options, *the spread tactic never turns out to be the worst tactic*, although it is the best one only if the underlying stock stabilizes. Tables 16-4 and 16-5 summarize the results the speculator could expect from invoking each of these five tactics. The tactics are:

1. Liquidate—sell the long put for a profit and do not reinvest.
2. Do nothing—continue to hold the long put.

TABLE 16-4.
Comparison of the five tactics.

By expiration, if XYZ . . .	the best strategy was . . .	and the worst strategy was . . .
Continues to fall dramatically	"Roll down"	Liquidate
Falls moderately further	Do nothing	Combine
Remains relatively unchanged	Spread	Combine or "roll down"
Rises moderately	Liquidate	"Roll down" or do nothing
Rises substantially	Combine	Do nothing

TABLE 16-5.
Results of adopting each of the five tactics.

XYZ Price at Expiration	"Roll Down" Profit	Do-Nothing Profit	Spread Profit	Liquidate Profit	Combine Profit
30	+ \$3,000 (B)	+\$1,800	+ \$500	+\$400 (W)	+ \$1,500
35	+ 2,000 (B)	+ 1,300	+ 500	+ 400 (W)	+ 1,000
41	+ 800 (B)	+ 700	+ 500	+ 400 (W)	+ 400
42	+ 600 (B)	+ 600 (B)	+ 500	+ 400	+ 300 (W)
43	+ 400	+ 500 (B)	+ 500 (B)	+ 400	+ 200 (W)
45	0 (W)	+ 300	+ 500 (B)	+ 400	0 (W)
46	0 (W)	+ 200	+ 400 (B)	+ 400 (B)	0 (W)
48	0 (W)	0 (W)	+ 200	+ 400 (B)	0 (W)
50	0	- 200 (W)	0	+ 400 (B)	0
54	0	- 200 (W)	0	+ 400 (B)	+ 400 (B)
60	0	- 200 (W)	0	+ 400	+ 1,000 (B)

3. “Roll down”—sell the long put, pocket the initial investment, and invest the remaining proceeds in out-of-the-money puts at a lower strike.
4. “Spread”—create a spread by selling the out-of-the-money put against the put already held.
5. “Combine”—create a combination by buying a call at a lower strike while continuing to hold the put.

Note that each tactic is the best one under one of the scenarios, but that the spread tactic is never the worst of the five. The actual results of each tactic, using the figures from the example above, are depicted in Table 16-5, where B denotes best tactic and W denotes worst one.

All the strategies are profitable if the underlying stock continues to fall dramatically, although the “roll down,” “do nothing,” and combinations work out best, because they continue to accrue profits if the stock continues to fall. Every time one takes partial profits, rolls down, or takes other measures, he is doing something bullish to his position. Those little bullish actions will be harmful if the underlying continues to decline. Rather, a trailing stop, placed *above* the declining stock price, might be the best tactic of all, because it allows one’s profits to run. If the underlying stock rises instead, only the combination outdistances the simplest tactic of all, liquidation.

If the underlying stock stabilizes, the “do-nothing” and “spread” tactics work out best. It would generally appear that the combination tactic or the “roll-down” tactic would be the most attractive, since neither one has any risk and both could generate large profits if the stock moved substantially. The advantage for the spread was substantial in call options, but in the case of puts, the premium received for the out-of-the-money put is not as large, and therefore the spread strategy loses some of its attractiveness. Finally, any of these tactics could be applied partially; for example, one could sell out half of a profitable long position in order to take some profits, and continue to hold the remainder.

LOSS-LIMITING ACTIONS

The foregoing discussion concentrated on how the put holder could retain or increase his profit. However, it is often the case in option buying that the holder of the option is faced with an unrealized loss. The put holder may also have several choices of action to take in this case. His first, and simplest, course of action would be to sell the put and take his loss. Although this is advisable in certain cases, especially when the underlying stock seems to have assumed a distinctly bullish stance, it is not always the wisest thing to do. The put holder who has a loss may also consider either “rolling up” to create a bearish spread or

entering into a calendar spread. Either of these actions could help him recover part or all of his loss.

THE “ROLLING-UP” STRATEGY

The reader may recall that a similar action to “rolling up,” termed “rolling down,” was available for call options held at a loss and was described in Chapter 3. The put buyer who owns a put at a loss may be able to create a spread that allows him to break even at a more favorable price at expiration. Such action will inevitably limit his profit potential, but is generally useful in recovering something from a put that might otherwise expire totally worthless.

Example: An investor initially purchases an XYZ October 45 put for 3 points when the underlying stock is at 45. However, the stock rises to 48 at a later date and the put that was originally bought for 3 points is now selling for 1.50 points. It is not unusual, by the way, for a put to retain this much of its value even though the stock has moved up and some amount of time has passed, since out-of-the-money puts tend to hold time value premium rather well. With XYZ at 48, an October 50 put might be selling for 3 points. The put holder could create a position designed to permit recovery of some of his losses by *selling two of the puts that he is long—October 45’s—and simultaneously buying one October 50 put*. The net cost for this transaction would be only commissions, since he receives \$300 from selling two puts at 1.50 each, which completely covers the \$300 cost of buying the October 50 put. The transactions are summarized in Table 16-6.

By selling 2 of the October 45 puts, the investor is now short an October 45 put. Since he also purchased an October 50 put, he has a spread (technically, a bear spread). He has spent no additional money, except commissions, to set up this spread, since the sale of the October 45’s covered the purchase of the October 50 put. *This strategy is most attractive when the debit involved to create the spread is small.* In this example, the debit is zero.

The effect of creating this spread is that *the investor has not increased his risk at all, but has raised the break-even point for his position*. That is, if XYZ merely falls a small distance, he will be able to get out even. Without the effect of creating the spread, the put holder would need XYZ to fall back to 42 at expiration in order for him to break even, since he originally paid 3 points for the October 45 put. His original risk was \$300. If XYZ continues to rise in price and the puts in the spread expire worthless, the net loss will still be only \$300 plus additional commissions. Admittedly, the commissions for the spread will increase the loss slightly, but they are small in comparison to the debit of the position (\$300). On the other hand, if the stock should fall back only slightly, to 47 by expiration,

TABLE 16-6.
Summary of rolling-up transactions.

Original trade:	Buy 1 October 45 put for 3 with XYZ at 45	\$300 debit
Later:	With XYZ at 48, sell 2 October 45's for 1.50 each and buy 1 October 50 put for 3	\$300 credit \$300 debit
Net position:	Long 1 October 50 put Short 1 October 45 put	\$300 debit

the spread will break even. At expiration, with XYZ at 47, the in-the-money October 50 put will be worth 3 points and the out-of-the-money October 45 put will expire worthless. Thus, the investor will recover his \$300 cost, except for commissions, with XYZ at 47 at expiration. His break-even point is raised from 42 to 47, a substantial improvement of his chances for recovery.

The implementation of this spread strategy reduces the profit potential of the position, however. The maximum potential of the spread is 2 points. If XYZ is anywhere below 45 at expiration, the spread will be worth 5 points, since the October 50 put will sell for 5 points more than the October 45 put. The investor has limited his potential profit to 2 points—the 5-point maximum width of the spread, less the 3 points that he paid to get into the position. He can no longer gain substantially on a large drop in price by the underlying stock. This is normally of little concern to the put holder faced with an unrealized loss and the potential for a total loss. He generally would be appreciative of getting out even or of making a small profit. The creation of the spread accomplishes this objective for him.

It should also be pointed out that he does not incur the maximum loss of his entire debit plus commissions, unless XYZ closes above 50 at expiration. If XYZ is anywhere below 50, the October 50 will have some value and the investor will be able to recover something from the position. This is distinctly different from the original put holding of the October 45, whereby the maximum loss would be incurred unless the stock were below 45 at expiration. *Thus, the introduction of the spread also reduces the chances of having to realize the maximum loss.*

In summary, the put holder faced with an unrealized loss may be able to create a spread by selling twice the number of puts that he is currently long and simultaneously buying the put at the next higher strike. This action should be used only if the spread can be transacted at a small debit or, preferably, at even money (zero debit). The spread

position offers a much better chance of breaking even and also reduces the possibility of having to realize the maximum loss in the position. However, the introduction of these loss-limiting measures reduces the maximum potential of the position if the underlying stock should subsequently decline in price by a significant amount. Using this spread strategy for puts would require a margin account, just as calls do.

THE CALENDAR SPREAD STRATEGY

Another strategy is sometimes available to the put holder who has an unrealized loss. If the put that he is holding has an intermediate-term or long-term expiration date, he might be able to create a *calendar spread* by selling the near-term put against the put that he currently holds.

Example: An investor bought an XYZ October 45 put for 3 points when the stock was at 45. The stock rises to 48, moving in the wrong direction for the put buyer, and his put falls in value to 1.50. He might, at that time, consider selling the near-term July 45 put for 1 point. The ideal situation would be for the July 45 put to expire worthless, reducing the cost of his long put by 1 point. Then, if the underlying stock declined below 45, he could profit after July expiration.

The major drawback to this strategy is that little or no profit will be made—in fact, a loss is quite possible—if the underlying stock falls back to 45 or below before the near-term July option expires. Puts display different qualities in their time value premiums than calls do, as has been noted before. With the stock at 45, the differential between the July 45 put and the October 45 put might not widen much at all. This would mean that the spread has not gained anything, and the spreader has a loss equal to his commissions plus the initial unrealized loss. In the example above, if XYZ dropped quickly back to 45, the July 45 might be worth 1.50 and the October worth 2.50. At this point, the spreader would have a loss on both sides of his spread: He sold the July 45 put for 1 and it is now 1.50; he bought the October 45 for 3 and it is now 2.50; plus he has spent two commissions to date and would have to spend two more to liquidate the position.

At this point, the strategist may decide to do nothing and take his chances that the stock will subsequently rally so that the July 45 put will expire worthless. However, if the stock continues to decline below 45, the spread will most certainly become more of a loss as both puts come closer to parity.

This type of spread strategy is not as attractive as the “rolling-up” strategy. In the “rolling-up” strategy, one is not subjected to a loss if the stock declines after the spread is established, although he does limit his profits. The fact that the calendar spread strategy can lead to a loss even if the stock declines makes it a less desirable alternative.

EQUIVALENT POSITIONS

Before considering other put-oriented strategies, the reader should understand the definition of an equivalent position. Two strategies, or positions, are equivalent when they have the same profit potential. They may have different collateral or investment requirements, but they have similar profit potentials. Many of the call-oriented strategies that were discussed in Part II of the book have an equivalent put strategy. One such case has already been described: The “protected short sale,” or shorting the common stock and buying a call, is equivalent to the purchase of a put. That is, both have a limited risk above the striking price of the option and relatively large profit potential to the downside. *An easy way to tell if two strategies are equivalent is to see if their profit graphs have the same shape.* The put purchase and the “protected short sale” have profit graphs with exactly the same shape (Figures 16-1 and 4-1, respectively). As more put strategies are discussed, it will always be mentioned if the put strategy is equivalent to a previously described call strategy. This may help to clarify the put strategies, which understandably may seem complex to the reader who is not familiar with put options.